



## Cloud boots

A patent filed by Apple entitled "Method and apparatus for administering the operating system of a net-booted environment" talks about how a network computer system comprising a server and various clients. Different clients can receive 'operating system' from the server with the help of one of the clients. Is this the beginning for an era by apple? May be yes. Apple has said that the OS "may also be downloaded as a computer program product, wherein the program may be transferred from a remote computer (e.g., a server) to a requesting computer (e.g., a client) by way of data signals embodied in a carrier wave or other propagation medium via a communication link (e.g., a modem or network connection)."

# Operating Systems

The potential of operating systems scaling in the future is huge. Here are some of the most promising technologies that will take OSes to a new level

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**B**y Although it might look like a bit distant, the possibility of every computer being booted over the cloud does not seem to be impossible by any means. A few months ago when Chrome CR-48 notebook was given away by Google for testing, quite a few loved the way everything was saved on the cloud and their android devices could also be used to fetch the same content. Those who unreliable connections faced severe problems with the device and were not-at-all impressed.

Future communication technologies worldwide and availability of service spreading at a rapid rate, it might be someday possible that we would all be booting our machines over a cloud hosted by our ISPs. With Apple featuring OS installation over Wi-Fi on its Macbook Air and Mac Mini product lines (<http://bit.ly/DairOS>) it would not be a difficult to imagine scenario where your netbook will be able to boot into its (at least, a secondary) operating system using your ISP's Data connection. With Apple recently acquiring patent for netbooting environment, it seems like the war for controlling the user's PC in more ways is going to go up and the day is not far when you would be receiving your OS over a network

connection. Certain relief for those of us who have been counting the number of hours wasted for OS installations.

Ever heard of parallel kernels? Lets consider a few facts about GPUs before we proceed. Graphics cards have as many as 1024 processors within them. On certain operations, they can beat CPUs by more than a scale of 20. A lot of algorithms (e.g. encryption algorithms) can run extremely fast on graphics processors when compared to CPU. And lastly a project called KGPU(<http://code.google.com/p/kgpu/>) is already trying to use CUDA to accelerate a few core parts of Linux kernel.

Taking all those into consideration, it would not be an unimaginable future with supercomputing capabilities in your computers ten years later since now being used every day to make your life isolated from all the hitch and glitch. Ten or fifteen years into the future, we might have our own weather forecaster, health assistant and a complete social planner on our machines, powered by the Operating system itself. By which we mean the 'graphics cards' thanks to parallel kernels.

The world today means 'RAM' when it talks 'memory' and means 'disks' when it talks 'storage'. Well, that might change quite a byte (well, that's 8 times a 'bit') in the coming future.

HP has been successful at creating the first prototype of a



## Semantic Desktop

A vision of a desktop which will be able to understand the meaning and importance of files stored on system. It will understand the needs and behaviour of its user and adapt the system accordingly. The operating systems would be able to use the files' data as metadata for search and interlink them with various other objects on the system. They will be able to analyse the behaviour of the user and predict his needs based on it. All this would be tightly integrated into the social and semantic web which would understand and exchange meanings of user's behaviour with desktop.

**Quick Links:** <http://goo.gl/abuKN>, <http://en.wikipedia.org/wiki/WinFS>, <http://www.w3.org/2001/sw/SW-FAQ>